Mutually Independent Hamiltonian Cycles of Pancake Networks

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Abstract

A hamiltonian cycle C of G is described as $\langle u_1, u_2, \ldots, u_{n(G)}, u_1\rangle$ to emphasize the order of nodes in C. Thus, u_1 is the beginning node and u_i is the i-th node in C. Two hamiltonian cycles of G beginning at a node x, $C_1 = \langle u_1, u_2, \ldots, u_{n(G)}, u_1\rangle$ and $C_2 = \langle v_1, v_2, \ldots, v_{n(G)}, v_1\rangle$, are independent if $x = u_1 = v_1$, and $u_i \neq v_i$ for every $2 \leq i \leq n(G)$. A set of hamiltonian cycles $\{C_1, C_2, \ldots, C_k\}$ of G are mutually independent if any two different hamiltonian cycles are independent. The mutually independent hamiltonicity of graph G, $IHC(G)$, is the maximum integer k such that for any node u of G there exist k-mutually independent hamiltonian cycles of G starting at u. In this paper, we are going to study $IHC(G)$ for the n-dimensional pancake graph P_n. We will prove that $IHC(P_3) = 1$ and $IHC(P_n) = n - 1$ if $n \geq 4$.

1 Introduction

An interconnection network connects the processors of parallel computers. Its architecture can be represented as a graph in which the nodes correspond to processors and the edges correspond to connections. Hence, we use graphs and networks interchangeably. There are many mutually conflicting requirements in designing the topology for computer networks. Akers and Krishnameurthy [1] proposed a family of interesting interconnection networks, the n-dimensional pancake graph P_n. They showed that the pancake graphs are node transitive. Hung et al. [7] studied the hamiltonian connectivity on the faulty pancake graphs. The embedding of cycles and trees into the pancake graphs where discussed in [3, 4, 7, 8]. Gates and Papadimitriou [5] studied the diameter of the pancake graphs. Up to now, we do not know the exact value of the diameter of the pancake graphs [6].

For a graph definitions and notation we follow [2]. $G = (V, E)$ is a *graph* if V is a finite set and E is a subset of $\{(u, v) \mid (u, v)$ is an unordered pair of $V\}$. We say that V is the *node set* and E is the *edge set*. We use $n(G)$ to denote $|V|$. Let S be a subset of V. The subgraph of G *induced* by S, $G[S]$, is the graph with $V(G[S]) = S$ and $E(G[S]) = \{(x, y) \mid (x, y) \in E(G)$ and $x, y \in S\}$. We use $G-S$ to denote the subgraph of G induced by $V - S$. Two nodes u and v are *adjacent* if (u, v) is an edge of G. The set of *neighbors* of u, denote by $N_G(u)$, is $\{v \mid (u, v) \in E\}$. The *degree* $d_G(u)$ of a node u of G is the number of edges incident with u. The *minimum degree* of G, written $\delta(G)$, is $\min\{d_G(x) \mid x \in V\}$. A *path* is a sequence of nodes represented by $\langle v_0, v_1, \ldots, v_k\rangle$ with no repeated node and (v_i, v_{i+1}) is an edge of G for all $0 \le i \le k-1$. We use $Q(i)$ to denote the i-th node v_i of $Q = \langle v_1, v_2, \ldots, v_k\rangle$. We also write the path $\langle v_0, v_1, \ldots, v_k\rangle$ as $\langle v_0, \ldots, v_i, Q, v_j, \ldots, v_k\rangle$, where Q is a path form v_i to v_j. A path is a *hamiltonian path* if it contains all nodes of G. A graph G is *hamiltonian connected* if there exists a hamiltonian path joining any two distinct nodes of G. A *cycle* is a path with at least three nodes such that the first node is the same as the last one. A *hamiltonian cycle* of G is a cycle that traverses every node of G. A graph is *hamiltonian* if it has a hamiltonian cycle.

A hamiltonian cycle C of graph G is described as $\langle u_1, u_2, \ldots, u_{n(G)}, u_1\rangle$ to emphasize the order of nodes in C. Thus, u_1 is the beginning node and u_i is the i-th node in C. Two hamiltonian cycles of G beginning at a node x, $C_1 = \langle u_1, u_2, \ldots, u_{n(G)}, u_1\rangle$ and $C_2 = \langle v_1, v_2, \ldots, v_{n(G)}, v_1\rangle$, are *independent* if $x = u_1 = v_1$, and $u_i \ne v_i$ for every $2 \le i \le n(G)$. A set of hamiltonian cycles $\{C_1, C_2, \ldots, C_k\}$ of G are *mutually independent* if any two different hamiltonian cycles are independent. The *mutually independent hamiltonianicity* of graph G, $IHC(G)$, is the maximum integer k such that for any node u of G there exist k-mutually independent hamiltonian cycles of G starting at u. Obviously, $IHC(G) \le \delta(G)$ if G is a hamiltonian graph.

The concept of mutually independent hamiltonian cycles can be applied in a lot of areas. Consider the following scenario. In Christmas, we have a holiday of 10 days. A tour agency will organize a 10-day tour to Italy. Suppose that there will be a lot of people joining this tour. However, the maximum number of people stay in each local area is limited, say 100 people, for the sake of hotel contract. One trivial solution is on the First-Come-First-Serve basis. So only 100 people can attend this tour. (Note that we can not schedule the tour in a pipelined manner because the holiday period is fixed.) Nonetheless, we observe that a tour is like a hamiltonian cycle based on a graph, in which a node is denoted as a hotel and any two nodes are joined with an edge if the associated two hotels can be traveled in a reasonable time. Therefore, we can organize all the different subgraphs, i.e. each subgraph has its own tour. In this way, we do not allow two subgroups stay in the same area during the same time period. In other words, any two different tours are indeed independent hamiltonian cycles. Suppose that there are 10-mutually independent hamiltonian cycles. Then we may allow 1000 people to visit Italy on Christmas vacation. For this reason, we would like to find the maximum number of mutually independent hamiltonian cycles. Such applications are useful for task scheduling and resource placement, which are also important for complier optimization to exploit parallelism.

In this paper, we study mutually independent hamiltonian cycles of pancake graph P_n. In the following section, we give the definition of the pancake graphs and review some of the previous work used in this paper. In section 3, we prove that $IHC(P_3) = 1$ and $IHC(P_n) = n - 1$ if $n \ge 4$.

2 The pancake graphs

Let n be a positive integer. We use $\langle n\rangle$ to denote the set $\{1, 2, \ldots, n\}$. The *n-dimensional pancake graph*, denoted by P_n, is a graph with the node set $V(P_n) = \{u_1u_2\ldots u_n \mid u_i \in \langle n\rangle$ and $u_i \ne u_j$ for $i \ne j\}$. The adjacency is defined as follows: $u_1u_2\ldots u_i\ldots u_n$ is adjacent to $v_1v_2\ldots v_i\ldots v_n$ through an edge of dimension i with $2 \le i \le n$ if $v_j = u_{i-j+1}$ for all $1 \le j \le i$ and $v_j = u_j$ for all $i < j \le n$. We will use boldface to denote a node of P_n. Hence, $\mathbf{u_1}, \mathbf{u_2}, \ldots, \mathbf{u_n}$ denote a sequence of nodes in P_n. In particular, $\mathbf{e}$ denotes the node $12\ldots n$. The pancake graphs P_2, P_3, and P_4 are illustrated in Figure 1.

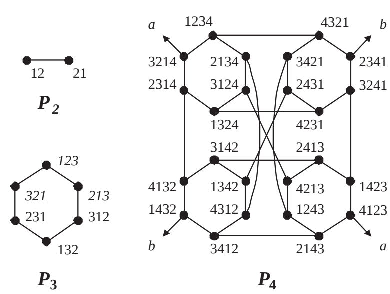

Figure 1: The pancake graphs P_2, P_3, and P_4.

By definition, P_n is an $(n-1)$-regular graph with $n!$ nodes. Moreover, it is node transitive [1]. Let $\mathbf{u} = u_1u_2\ldots u_n$ be an arbitrary node of P_n. We use $(\mathbf{u})_i$ to denote the i-th component u_i of $\mathbf{u}$, and use $P_n^{\{i\}}$ to denote the i-th subgraph of P_n induced by those nodes $\mathbf{u}$ with $(\mathbf{u})_n = i$. Then P_n can be decomposed into n node disjoint subgraphs $P_n^{\{i\}}$ for all $i \le n$ such that each $P_n^{\{i\}}$ is isomorphic to P_{n-1}. Thus, the pancake graph can be constructed recursively. Let $H \subseteq \langle n \rangle$, we use P_n^H to denote the subgraph of P_n induced by $\cup_{i \in H} V(P_n^{\{i\}})$. For $1 \leq i \neq j \leq n$, we use $P_n^{\{i,j\}}$ to denote the subgraph of P_n induced by those nodes $\mathbf{u}$ with $(\mathbf{u})_{n-1} = i$ and $(\mathbf{u})_n = j$. Obviously, $P_n^{\{i,j\}} \neq P_n^{\{j,i\}}$ and $P_n^{\{i,j\}}$ is isomorphic to P_{n-2}. By definition, there is exactly one neighbor $\mathbf{v}$ of $\mathbf{u}$ such that $\mathbf{u}$ and $\mathbf{v}$ are adjacent through an i-dimensional edge with $2 \leq i \leq n$. We use $(\mathbf{u})^i$ to denote the unique i-neighbor of $\mathbf{u}$. We have $((\mathbf{u})^i)^i = \mathbf{u}$ and $(\mathbf{u})^n \in P_n^{\{(\mathbf{u})_1\}}$. For $1 \leq i, j \leq n$ and $i \neq j$, we use $E^{i,j}$ to denote the set of edges between $P_n^{\{i\}}$ and $P_n^{\{j\}}$.

Lemma 1. *Let $i, j \in \langle n \rangle$ with $i \neq j$ and $n \geq 3$. Then $|E_n^{i,j}| = (n-2)!$.*

Lemma 2. *Let $\mathbf{u}$ and $\mathbf{v}$ be two distinct nodes of P_n with $d(\mathbf{u}, \mathbf{v}) \leq 2$. Then $(\mathbf{u})_1 \neq (\mathbf{v})_1$.*

Theorem 1. *[7] Suppose that F is a subset of $V(P_n)$ with $|F| \leq n-4$ and $n \geq 4$. Then $P_n - F$ is hamiltonian connected.*

Theorem 2. *Let $\{a_1, a_2, \ldots, a_r\}$ be a subset of $\langle n \rangle$ for some positive integer $r \in \langle n \rangle$ with $n \geq 5$. Assume that $\mathbf{u}$ and $\mathbf{v}$ are two distinct nodes of P_n with $\mathbf{u} \in P_n^{\{a_1\}}$ and $\mathbf{v} \in P_n^{\{a_r\}}$. Then there is a hamiltonian path $H = \langle \mathbf{u} = \mathbf{x_1}, H_1, \mathbf{y_1}, \mathbf{x_2}, H_2, \mathbf{y_2}, \ldots, \mathbf{x_r}, H_r, \mathbf{y_r} = \mathbf{v} \rangle$ of $\cup_{i=1}^{r} P_n^{\{a_i\}}$ joining $\mathbf{u}$ to $\mathbf{v}$ such that $\mathbf{x_1} = \mathbf{u}$, $\mathbf{y_r} = \mathbf{v}$, and H_i is a hamiltonian path of $P_n^{\{a_i\}}$ joining $\mathbf{x_i}$ to $\mathbf{y_i}$ for every $1 \leq i \leq r$.*

Proof. We set $\mathbf{x_1}$ as $\mathbf{u}$ and set that $\mathbf{y_r}$ as $\mathbf{v}$. We know that $P_n^{\{a_i\}}$ is isomorphic to P_{n-1} for every $i \in \langle r \rangle$. By Theorem 1, this statement holds for $r = 1$. Thus, we assume that $r \geq 2$. By Lemma 1, $|E_n^{a_i, a_{i+1}}| = (n-2)! \geq 6$ for every $i \in \langle r-1 \rangle$. We choose $(\mathbf{y_i}, \mathbf{x_{i+1}}) \in E^{a_i, a_{i+1}}$ for every $i \in \langle r-1 \rangle$ with $\mathbf{y_1} \neq \mathbf{x_1}$ and $\mathbf{x_r} \neq \mathbf{y_r}$. By Theorem 1, there is a hamiltonian path H_i of $P_n^{\{a_i\}}$ joining $\mathbf{x_i}$ to $\mathbf{y_i}$ for every $i \in \langle r \rangle$. Then $H = \langle \mathbf{u} = \mathbf{x_1}, H_1, \mathbf{y_1}, \mathbf{x_2}, H_2, \mathbf{y_2}, \ldots, \mathbf{x_r}, H_r, \mathbf{y_r} = \mathbf{v} \rangle$ forms a desired path. See Figure 2 for illustration on P_n. $\square$

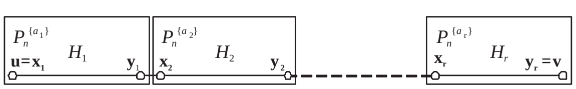

Figure 2: Illustration for Theorem 2 on P_n.

3 $IHC(P_n)$

Lemma 3. *Let $k \in \langle n \rangle$ with $n \geq 4$, and let $\mathbf{x}$ be a node of P_n. There is a hamiltonian path P of $P_n - \{\mathbf{x}\}$ joining the node $(\mathbf{x})^n$ to some node $\mathbf{v}$ with $(\mathbf{v})_1 = k$.*

Proof. Suppose that $n = 4$. Since P_4 is node transitive, we may assume that $\mathbf{x} = 1234$. The required paths of $P_4 - \{1234\}$ are listed below:

$k = 1$ (4321)(3421)(2431)(4231)(1324)(3124)(2134)(4312) (1342)(3124)(4132)(2314)(3214)(4123)(2143)(3412) (1432)(2341)(3241)(1423)(2413)(4213)(1243)
$k = 2$ (4321)(3421)(2431)(4231)(1324)(3124)(2134)(4312) (1342)(3142)(2413)(4213)(1243)(2143)(3412)(1432) (4132)(2314)(3214)(4123)(1423)(3241)(2341)
$k = 3$ (4321)(3421)(2431)(4231)(1324)(3124)(2134)(4312) (1342)(3142)(4132)(2314)(3214)(4123)(1423)(2413) (4213)(1243)(2143)(3412)(1432)(2341)(3241)
$k = 4$ (4321)(3421)(2431)(1342)(3142)(4132)(2314)(3214) (4123)(2143)(1243)(4213)(2413)(1423)(3241)(2341) (1432)(3412)(4312)(2134)(3124)(1324)(4231)

With Theorem 1, we can find the required hamiltonian path in P_n for every $n \geq 5$. $\square$

Lemma 4. *Let $a, b \in \langle n \rangle$ with $a \neq b$ and $n \geq 4$, and let $\mathbf{x}$ be a node of P_n. There is a hamiltonian path P of $P_n - \{\mathbf{x}\}$ joining a node $\mathbf{u}$ with $(\mathbf{u})_1 = a$ to a node $\mathbf{v}$ with $(\mathbf{v})_1 = b$.*

Proof. Suppose that $n = 4$. Since P_4 is node transitive, we may assume that $\mathbf{x} = 1234$. Without loss of generality, we may assume that $a < b$. We have $a \neq 4$ and $b \neq 1$. The required paths of $P_4 - \{1234\}$ are listed below:

$a = 1$ and $b = 2$ (1423)(4123)(3214)(2314)(1324)(3124)(4213)(2413) (3142)(4132)(1432)(3412)(2143)(1243)(3421)(4321) (2341)(3241)(4231)(2431)(1342)(4312)(2134)
$a = 1$ and $b = 3$ (1423)(4123)(2143)(1243)(4213)(2413)(3142)(1342) (2431)(3421)(4321)(2341)(3241)(4231)(1324)(3124) (2134)(4312)(3412)(1432)(4132)(2314)(3214)
$a = 1$ and $b = 4$ (1423)(2413)(3142)(1342)(2431)(3421)(4321)(2341) (3241)(4231)(1324)(2314)(3214)(4123)(2143)(1243) (4213)(3124)(2134)(4312)(3412)(1432)(4132)
$a = 2$ and $b = 3$ (2134)(4312)(1342)(3142)(2413)(4213)(1243)(2143) (3412)(1432)(4132)(2314)(3214)(4123)(1423)(3241) (2341)(4321)(3421)(2431)(4231)(1324)(3124)
$a = 2$ and $b = 4$ (2134)(3124)(1324)(2314)(3214)(4123)(2143)(1243) (4213)(2413)(1423)(3241)(4231)(2431)(3421)(4321) (2341)(1432)(3412)(4312)(1342)(3142)(4132)
$a = 3$ and $b = 4$ (3214)(4123)(2143)(1243)(4213)(3124)(2134)(4312) (3412)(1432)(2341)(4321)(3421)(2431)(1342)(3142) (2413)(1423)(3241)(4231)(1324)(2314)(4132)

With Theorem 1, we can find the required hamiltonian path on P_n for every $n \geq 5$. $\square$

Lemma 5. *Let $n \geq 4$, and $a, b \in \langle n \rangle$ with $a \neq b$. Assume that $\mathbf{x}$ and $\mathbf{y}$ are two adjacent nodes of P_n. There is a hamiltonian path P of $P_n - \{\mathbf{x}, \mathbf{y}\}$ joining a node $\mathbf{u}$ with $(\mathbf{u})_1 = a$ to a node $\mathbf{v}$ with $(\mathbf{v})_1 = b$.*

Proof. Since P_n is node transitive, we may assume that $\mathbf{x} = \mathbf{e}$ and $\mathbf{y} = (\mathbf{e})^i$ for some $i \in \{2, 3, \ldots, n\}$. Without loss of generality, we assume that $a < b$. Thus, $a \neq n$ and $b \neq 1$. We prove this lemma by induction on n. For $n = 4$, the required paths of $P_4 - \{1234, (1234)^i\}$ are listed below:

$\mathbf{y} = 2134$
$a = 1$ and $b = 2$ (1432)(2413)(3142)(4132)(1432)(3412)(4312)(1342) (2431)(3421)(4321)(2341)(3241)(4231)(1324)(3124) (4213)(1243)(2143)(4123)(3214)(2314)
$a = 1$ and $b = 3$ (1432)(4123)(2143)(1243)(3421)(4321)(2341)(3241) (4231)(2431)(1342)(4312)(3412)(1432)(4132)(3142) (2413)(4213)(3124)(1324)(2314)(3214)
$a = 1$ and $b = 4$ (1423)(4123)(3214)(2314)(1324)(3124)(4213)(2413) (3142)(4132)(1432)(2341)(3241)(4231)(2431)(1342) (4312)(3412)(2143)(1243)(3421)(4321)
$a = 2$ and $b = 3$ (2314)(3214)(4123)(2143)(1243)(4213)(3124)(1324) (4231)(2431)(1342)(4312)(3412)(1432)(4132)(3142) (2413)(1423)(3241)(2341)(4321)(3421)
$a = 2$ and $b = 4$ (2314)(3214)(4123)(2143)(3412)(4312)(1342)(2431) (3421)(1243)(4213)(3124)(1324)(4231)(3241)(1423) (2413)(3142)(4132)(1432)(2341)(4321)
$a = 3$ and $b = 4$ (3214)(4123)(2143)(1243)(3421)(2431)(4231)(3241) (1423)(2413)(4213)(3124)(1324)(2314)(4132)(3142) (1342)(4312)(3412)(1432)(2341)(4321)

$\mathbf{y} = 3214$
$a = 1$ and $b = 2$ (1423)(4123)(2143)(1243)(3421)(4321)(2341)(3241) (4231)(2431)(1342)(3142)(2413)(4213)(3124)(1324) (2314)(4132)(1432)(3412)(4312)(2134)
$a = 1$ and $b = 3$ (1423)(4123)(2143)(1243)(4213)(2413)(3142)(1342) (2431)(3421)(4321)(2341)(3241)(4231)(1324)(2314) (4132)(1432)(3412)(4312)(2134)(3124)
$a = 1$ and $b = 4$ (1423)(4123)(2143)(1243)(3421)(2431)(1342)(3142) (2413)(4213)(3124)(2134)(4312)(3412)(1432)(4132) (2314)(1324)(4231)(3241)(2341)(4321)
$a = 2$ and $b = 3$ (2134)(4312)(1342)(2431)(4231)(3241)(1423)(4123) (2143)(3412)(1432)(2341)(4321)(3421)(1243)(4213) (2413)(3142)(4132)(2314)(1324)(3124)
$a = 2$ and $b = 4$ (2134)(3124)(4213)(2413)(3142)(1342)(4312)(3412) (1432)(4132)(2314)(1324)(4231)(2431)(3421)(1243) (2143)(4123)(1423)(3241)(2341)(4321)
$a = 3$ and $b = 4$ (3124)(2134)(4312)(1342)(3142)(2413)(4213)(1243) (3421)(2431)(4231)(1324)(2314)(4132)(1432)(3412) (2143)(4123)(1423)(3241)(2341)(4321)

$\mathbf{y} = 4321$
$a = 1$ and $b = 2$ (1423)(4123)(3214)(2314)(4132)(3142)(2413)(4213) (3124)(1324)(4231)(3241)(2341)(1432)(3412)(2143) (1243)(3421)(2431)(1342)(4312)(2134)
$a = 1$ and $b = 3$ (1423)(4123)(2143)(3412)(1432)(2341)(3241)(4231) (1324)(3124)(2134)(4312)(1342)(2431)(3421)(1243) (4213)(2413)(3142)(4132)(2314)(3214)
$a = 1$ and $b = 4$ (1423)(2413)(4213)(3124)(2134)(4312)(3412)(2143) (1243)(3421)(2431)(1342)(3142)(4132)(1432)(2341) (3241)(4231)(1324)(2314)(3214)(4123)
$a = 2$ and $b = 3$ (2134)(4312)(1342)(3142)(4132)(2314)(3214)(4123) (2143)(3412)(1432)(2341)(3241)(1423)(2413)(4213) (1243)(3421)(2431)(4231)(1324)(3124)
$a = 2$ and $b = 4$ (2134)(3124)(4213)(2413)(1423)(3241)(2341)(1432) (4132)(3142)(1342)(4312)(3412)(2143)(1243)(3421) (2431)(4231)(1324)(2314)(3214)(4123)
$a = 3$ and $b = 4$ (3214)(2314)(1324)(4231)(3241)(2341)(1432)(4132) (3142)(1342)(2431)(3421)(1243)(2143)(3412)(4312) (2134)(3124)(4213)(2413)(1423)(4123)

Suppose that this lemma is true for P_k for every $4 \leq k < n$. We have the following cases:

Case 1. $\mathbf{y} = (\mathbf{e})^i$ for some $i \neq 1$ and $i \neq n$, i. e. $\mathbf{y} \in P_n^{\{n\}}$. Let $c \in \langle n-1 \rangle - \{a\}$. By induction, there is a hamiltonian path R of $P_n^{\{n\}} - \{\mathbf{e}, (\mathbf{e})^i\}$ joining a node $\mathbf{u}$ with $(\mathbf{u})_1 = a$ to a node $\mathbf{z}$ with $(\mathbf{z})_1 = c$. We choose a node $\mathbf{v}$ in $P_n^{\langle n-1 \rangle - \{c\}}$ with $(\mathbf{v})_1 = b$. By Theorem 2, there is a hamiltonian path H of $P_n^{\langle n-1 \rangle}$ joining the node $(\mathbf{z})^n$ to $\mathbf{v}$. Then $\langle \mathbf{u}, R, \mathbf{z}, (\mathbf{z})^n, H, \mathbf{v} \rangle$ forms the desired path.

Case 2. $\mathbf{y} = (\mathbf{e})^n$, i. e. $\mathbf{y} \in P_n^{\{1\}}$. Let $c \in \langle n-1 \rangle - \{1, a\}$ and $d \in \langle n-1 \rangle - \{1, b, c\}$. By Lemma 4, there is a hamiltonian path R of $P_n^{\{n\}} - \{\mathbf{e}\}$ joining a node $\mathbf{u}$ with $(\mathbf{u})_1 = a$ to a node $\mathbf{w}$ with $(\mathbf{w})_1 = c$. Again, there is a hamiltonian path H of $P_n^{\{1\}} - \{(\mathbf{e})^n\}$ joining a node $\mathbf{z}$ with $(\mathbf{z})_1 = d$ to a node $\mathbf{v}$ with $(\mathbf{v})_1 = b$. By Theorem 2, there is a hamiltonian path Q of $P_n^{\langle n-1 \rangle - \{1\}}$ joining the node $(\mathbf{w})^n$ to the node $(\mathbf{z})^n$. Then $\langle \mathbf{u}, R, \mathbf{w}, (\mathbf{w})^n, Q, (\mathbf{z})^n, \mathbf{z}, H, \mathbf{v} \rangle$ forms the desired path. □

Lemma 6. *Let $a, b \in \langle n \rangle$ with $n \geq 4$. Assume that $\mathbf{x}$ is a node of P_n, and $\mathbf{x_1}$ and $\mathbf{x_2}$ are two distinct neighbors of $\mathbf{x}$. There is a hamiltonian path P of $P_n - \{\mathbf{x}, \mathbf{x_1}, \mathbf{x_2}\}$ joining a node $\mathbf{u}$ with $(\mathbf{u})_1 = a$ to a node $\mathbf{v}$ with $(\mathbf{v})_1 = b$.*

Proof. Since P_n is node transitive, we may assume that $\mathbf{x} = \mathbf{e}$. Moreover, we assume that $\mathbf{x_1} = (\mathbf{e})^i$ and $\mathbf{x_2} = (\mathbf{e})^j$ for some $i, j \in \langle n \rangle - \{1\}$ with $i < j$. Without loss of generality, we assume that $a < b$. Thus, $a \neq n$ and $b \neq 1$. We prove this lemma by induction on n. For $n = 4$, the required paths of $P_4 - \{1234, (1234)^i, (1234)^j\}$ are listed below:

$\mathbf{x_1} = 2134$ and $\mathbf{x_2} = 3214$
$a = 1$ and $b = 2$ (1423)(4123)(2143)(1243)(3421)(4321)(2341) (3241)(4231)(2431)(1342)(4312)(3412)(1432) (4132)(3142)(2413)(4213)(3124)(1324)(2314)
$a = 1$ and $b = 3$ (1423)(4123)(2143)(1243)(3421)(4321)(2341) (3241)(4231)(2431)(1342)(4312)(3412)(1432) (4132)(2314)(1324)(3124)(4213)(2413)(3142)
$a = 1$ and $b = 4$ (1423)(4123)(2143)(1243)(3421)(4321)(2341) (3241)(4231)(2431)(1342)(3142)(2413)(4213) (3124)(1324)(2314)(4132)(1432)(3412)(4312)
$a = 2$ and $b = 3$ (2143)(4123)(1423)(3241)(4231)(2431)(1342) (4312)(3412)(1432)(2341)(4321)(3421)(1243) (4213)(2413)(3142)(4132)(2314)(1324)(3124)
$a = 2$ and $b = 4$ (2143)(4123)(1423)(2413)(3142)(1342)(4312) (3412)(1432)(4132)(2314)(1324)(3124)(4213) (1243)(3421)(2431)(4231)(3241)(2341)(4321)
$a = 3$ and $b = 4$ (3124)(4213)(2413)(3142)(1342)(4312)(3412) (1432)(4132)(2314)(1324)(4231)(2431)(3421) (1243)(2143)(4123)(1423)(3241)(2341)(4321)

$\mathbf{x_1} = 2134$ and $\mathbf{x_2} = 4321$
$a = 1$ and $b = 2$ (1423)(2413)(3142)(4132)(1432)(2341)(3241) (4231)(1324)(3124)(4213)(1243)(3421)(2431) (1342)(4312)(3412)(2143)(4123)(3214)(2314)
$a = 1$ and $b = 3$ (1423)(4123)(2143)(1243)(3421)(2431)(1342) (4312)(3412)(1432)(2341)(3241)(4231)(1324) (3124)(4213)(2413)(3142)(4132)(2314)(3214)
$a = 1$ and $b = 4$ (1423)(4123)(3214)(2314)(1324)(3124)(4213) (2413)(3142)(1342)(4312)(3412)(2143)(1243) (3421)(2431)(4231)(3241)(2341)(1432)(4132)

$\mathbf{x_1} = 2134$ and $\mathbf{x_2} = 4321$
$a = 2$ and $b = 3$ (2314)(3214)(4123)(2143)(3412)(4312)(1342) (3142)(4132)(1432)(2341)(3241)(1423)(2413) (4213)(1243)(3421)(2431)(4231)(1324)(3124)
$a = 2$ and $b = 4$ (2314)(3214)(4123)(2143)(1243)(3421)(2431) (4231)(1324)(3124)(4213)(2413)(1423)(3241) (2341)(1432)(3412)(4312)(1342)(3142)(4132)
$a = 3$ and $b = 4$ (3214)(2314)(4132)(3142)(1342)(4312)(3412) (1432)(2341)(3241)(1423)(2413)(4213)(3124) (1324)(4231)(2431)(3421)(1243)(2143)(4123)

$\mathbf{x_1} = 3214$ and $\mathbf{x_2} = 4321$
$a = 1$ and $b = 2$ (1423)(4123)(2143)(1243)(3421)(2431)(1342) (4312)(3412)(1432)(2341)(3241)(4231)(1324) (2314)(4132)(3142)(2413)(4213)(3124)(2134)
$a = 1$ and $b = 3$ (1423)(4123)(2143)(3412)(1432)(2341)(3241) (4231)(1324)(2314)(4132)(3142)(2413)(4213) (1243)(3421)(2431)(1342)(4312)(2134)(3124)
$a = 1$ and $b = 4$ (1423)(2413)(4213)(3124)(2134)(4312)(3412) (1432)(2341)(3241)(4231)(1324)(2314)(4132) (3142)(1342)(2431)(3421)(1243)(2143)(4123)
$a = 2$ and $b = 3$ (2134)(4312)(3412)(1432)(2341)(3241)(4231) (1324)(2314)(4132)(3142)(1342)(2431)(3421) (1243)(2143)(4123)(1423)(2413)(4213)(3124)
$a = 2$ and $b = 4$ (2134)(3124)(4213)(2413)(3142)(1342)(4312) (3412)(1432)(2341)(3241)(1423)(4123)(2143) (1243)(3421)(2431)(4231)(1324)(2314)(4132)
$a = 3$ and $b = 4$ (3124)(2134)(4312)(3412)(1432)(2341)(3241) (4231)(1324)(2314)(4132)(3142)(1342)(2431) (3421)(1243)(2143)(4123)(1423)(2413)(4213)

Suppose that this lemma is true for P_k for every $4 \le k < n$. We have the following cases:

Case 1. $j \neq n$, i. e. $\mathbf{x_1} \in P_n^{\{n\}}$ and $\mathbf{x_2} \in P_n^{\{n\}}$. Let $c \in \langle n-1 \rangle - \{1, a\}$. By induction, there is a hamiltonian path R of $P_n^{\{n\}} - \{\mathbf{e}, \mathbf{x_1}, \mathbf{x_2}\}$ joining a node $\mathbf{u}$ with $(\mathbf{u})_1 = a$ to a node $\mathbf{z}$ with $(\mathbf{z})_1 = c$. We choose a node $\mathbf{v}$ in $P_n^{\{1\}}$ with $(\mathbf{v})_1 = b$. By Theorem 2, there is a hamiltonian path H of $P_n^{\langle n-1 \rangle}$ joining the node $(\mathbf{z})^n$ to $\mathbf{v}$. We set $P = \langle \mathbf{u}, R, \mathbf{z}, (\mathbf{z})^n, H, \mathbf{v}, \rangle$. Then P forms the desired path.

Case 2. $j = n$, i. e. $\mathbf{x_1} \in P_n^{\{n\}}$ and $\mathbf{x_2} \in P_n^{\{1\}}$. Let $c \in \langle n-1 \rangle - \{1, a\}$ and $d \in \langle n-1 \rangle - \{1, b, c\}$. By Lemma 5, there is a hamiltonian path R of $P_n^{\{n\}} - \{\mathbf{e}, \mathbf{x_1}\}$ joining a node $\mathbf{u}$ with $(\mathbf{u})_1 = a$ to a node $\mathbf{z}$ with $(\mathbf{z})_1 = c$. By Lemma 4, there is a hamiltonian path H of $P_n^{\{1\}} - \{\mathbf{x_2}\}$ joining a node $\mathbf{w}$ with $(\mathbf{w})_1 = d$ to a node $\mathbf{v}$ with $(\mathbf{v})_1 = b$. By Theorem 2, there is a hamiltonian Q of $P_n^{\langle n-1 \rangle - \{1\}}$ joining the node $(\mathbf{z})^n$ to the node $(\mathbf{w})^n$. We set $P = \langle \mathbf{u}, R, \mathbf{z}, (\mathbf{z})^n, Q, (\mathbf{w})^n, \mathbf{w}, H, \mathbf{v}, \rangle$. Then P forms the desired path. $\square$

Our main result for the pancake graph P_n is stated in the following theorem.

Theorem 3. *$IHC(P_3) = 1$ and $IHC(P_n) = n-1$ if $n \ge 4$.*

Proof. It is easy to see that P_3 is isomorphic to a cycle with six nodes, so $IHC(P_3) = 1$. P_n is $(n-1)$-regular graph, it is clear that $IHC(P_n) \le n-1$. Since P_n is node transitive, we only need to show that there exist $(n-1)$-mutually independent hamiltonian cycles of P_n starting form the node $\mathbf{e}$. For $n = 4$, we prove that $IHC(P_4) \ge 3$ by listing the required hamiltonian cycles as following:

C_1	(1234)(2134)(4312)(3412)(2143)(1243)(4213)(3124)(1324) (4231)(3241)(2341)(1432)(4132)(2314)(3214)(4123)(1423) (2413)(3142)(1342)(2431)(3421)(4321)(1234)
C_2	(1234)(3214)(2314)(1324)(3124)(4213)(2413)(1423)(4123) (2143)(1243)(3421)(4321)(2341)(3241)(4231)(2431)(1342) (3142)(4132)(1432)(3412)(4312)(2134)(1234)
C_3	(1234)(4321)(2341)(1432)(4132)(2314)(1324)(4231)(3241) (1423)(2413)(3142)(1342)(2431)(3421)(1243)(4213)(3124) (2134)(4312)(3412)(2143)(4123)(3214)(1234)

Let $n \ge 5$. Let B be the $(n-1) \times n$ matrix with

$$b_{i,j} = \begin{cases} i+j-1 & \text{if } i+j-1 \le n, \\ i+j-n+1 & \text{if } n < i+j-1. \end{cases}$$

More precisely,

$$B = \begin{bmatrix} 1 & 2 & 3 & 4 & \cdots & n-1 & n \\ 2 & 3 & 4 & 5 & \cdots & n & 1 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ n-1 & n & 1 & 2 & \cdots & n-3 & n-2 \end{bmatrix}.$$

It is not hard to see that $(b_{i,1}, b_{i,2}, \ldots, b_{i,n})$ forms a permutation of $\{1, 2, \ldots, n\}$ for every i with $1 \le i \le n-1$. Moreover, $b_{i,j} \neq b_{i',j}$ for any $1 \le i < i' \le n-1$ and $1 \le j \le n$. In other words, B forms a latin rectangle with entries in $\{1, 2, \ldots, n\}$.

We construct $\{C_1, C_2, \ldots, C_{n-1}\}$ as follows:

(**1**) $k = 1$. By Lemma 3, there is a hamiltonian path H_1 of $P_n^{\{b_{1,n}\}} - \{\mathbf{e}\}$ joining a node $\mathbf{x}$ with $\mathbf{x} \neq (\mathbf{e})^{n-1}$ and $(\mathbf{x})_1 = n-1$ to the node $(\mathbf{e})^{n-1}$. By Theorem 2, there is a hamiltonian path H_2 of $\cup_{t=1}^{n-1} P_n^{\{b_{1,t}\}}$ joining the node $(\mathbf{e})^n$ to the node $(\mathbf{x})^n$ with $H_2(i + (j-1)(n-1)!) \in P_n^{\{b_{1,j}\}}$ for every $i \in \langle (n-1)! \rangle$ and for every $j \in \langle n-1 \rangle$. We set $C_1 = \langle \mathbf{e}, (\mathbf{e})^n, H_2, (\mathbf{x})^n, \mathbf{x}, H_1, (\mathbf{e})^{n-1}, \mathbf{e} \rangle$.

(**2**) $k = 2$. By Lemma 5, there is a hamiltonian path Q_1 of $P_n^{\{b_{2,n-1}\}} - \{\mathbf{e}, (\mathbf{e})^2\}$ joining a node $\mathbf{y}$ with $(\mathbf{y})_1 = n-1$ to a node $\mathbf{z}$ with $(\mathbf{z})_1 = 1$. By Theorem 2, there is a hamiltonian Q_2 of $\cup_{t=1}^{n-2} P_n^{\{b_{2,t}\}}$ joining the node $((\mathbf{e})^2)^n$ to the node $(\mathbf{y})^n$ such that $Q_2(i + (j-1)(n-1)!) \in P_n^{\{b_{2,j}\}}$ for every $i \in \langle (n-1)! \rangle$ and for every $j \in \langle n-2 \rangle$. By Theorem 1, there is a hamiltonian path Q_3 of $P_n^{\{b_{2,n}\}}$ joining the node $(\mathbf{z})^n$ to the node $(\mathbf{e})^n$. We set $C_2 = \langle \mathbf{e}, (\mathbf{e})^2, ((\mathbf{e})^2)^n, Q_2, (\mathbf{y})^n, \mathbf{y}, Q_1, \mathbf{z}, (\mathbf{z})^n, Q_3, (\mathbf{e})^n, \mathbf{e} \rangle$.

(**3**) $3 \le k \le n-1$. By Lemma 6, there is a hamiltonian path R_1^k of $P_n^{\{b_{k,n-k+1}\}} - \{\mathbf{e}, (\mathbf{e})^{k-1}, (\mathbf{e})^k\}$ joining a node $\mathbf{w_k}$ with $(\mathbf{w_k})_1 = n-1$ to a node $\mathbf{v_k}$ with $(\mathbf{v_k})_1 = 1$. By Theorem 2, there is a hamiltonian path R_2^k of $\cup_{t=1}^{n-k} P_n^{\{b_{k,t}\}}$ joining the node $((\mathbf{e})^k)^n$ to the node $(\mathbf{w_k})^n$ such that $R_2^k(i + (j-1)(n-1)!) \in P_n^{\{b_{k,j}\}}$ for every $i \in \langle (n-1)! \rangle$ and for every $j \in \langle n-k \rangle$. Again, there is a hamiltonian path R_3^k of $\cup_{t=n-k+2}^{n} P_n^{\{b_{k,t}\}}$ joining the node $(\mathbf{v_k})^n$ to the node $((\mathbf{e})^{k-1})^n$ such that $R_3^k(i+(j-1)(n-1)!) \in P_n^{\{b_{k,n-k+j+1}\}}$ for every $i \in \langle (n-1)! \rangle$ and for every $j \in \langle k-1 \rangle$. We set $C_k = \langle \mathbf{e}, (\mathbf{e})^k, ((\mathbf{e})^k)^n, R_2^k, (\mathbf{w_k})^n, \mathbf{w_k}, R_1^k, \mathbf{v_k}, (\mathbf{v_k})^n, R_3^k, ((\mathbf{e})^{k-1})^n, (\mathbf{e})^{k-1}, \mathbf{e} \rangle$.

Then $\{C_1, C_2, \ldots, C_{n-1}\}$ forms a set of $(n-1)$-mutually independent hamiltonian cycles of P_n starting from the node $\mathbf{e}$. $\square$

Example: We illustrate the proof of Theorem 3 with $n = 5$ as follows:

We set

$$B = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 5 & 1 \\ 3 & 4 & 5 & 1 & 2 \\ 4 & 5 & 1 & 2 & 3 \end{bmatrix}.$$

Then we construct $\{C_1, C_2, C_3, C_4\}$ as follows:

(**1**) $k = 1$. By Lemma 3, there is a hamiltonian path H_1 of $P_5^{\{b_{1,5}\}} - \{\mathbf{e}\}$ joining a node $\mathbf{x}$ with $\mathbf{x} \neq (\mathbf{e})^4$ and $(\mathbf{x})_1 = 4$ to the node $(\mathbf{e})^4$. By Theorem 2, there is a hamiltonian path H_2 of $\cup_{t=1}^{4} P_5^{\{b_{1,t}\}}$ joining the node $(\mathbf{e})^5$ to the node $(\mathbf{x})^5$ with $H_2(i + 24(j-1)) \in P_5^{\{b_{1,j}\}}$ for every $i \in \langle 24 \rangle$ and for every $j \in \langle 4 \rangle$. We set $C_1 = \langle \mathbf{e}, (\mathbf{e})^5, H_2, (\mathbf{x})^5, \mathbf{x}, H_1, (\mathbf{e})^4, \mathbf{e} \rangle$.

(**2**) $k = 2$. By Lemma 5, there is a hamiltonian path Q_1 of $P_5^{\{b_{2,4}\}} - \{\mathbf{e}, (\mathbf{e})^2\}$ joining a node $\mathbf{y}$ with $(\mathbf{y})_1 = 4$ to a node $\mathbf{z}$ with $(\mathbf{z})_1 = 1$. By Theorem 2, there is a hamiltonian Q_2 of $\cup_{t=1}^{3} P_5^{\{b_{2,t}\}}$ joining the node $((\mathbf{e})^2)^5$ to the node $(\mathbf{y})^5$ such that $Q_2(i + 24(j-1)) \in P_5^{\{b_{2,j}\}}$ for every $i \in \langle 24 \rangle$ and for every $j \in \langle 3 \rangle$. By Theorem 1, there is a hamiltonian path Q_3 of $P_5^{\{b_{2,5}\}}$ joining the node $(\mathbf{z})^5$ to the node $(\mathbf{e})^5$. We set $C_2 = \langle \mathbf{e}, (\mathbf{e})^2, ((\mathbf{e})^2)^5, Q_2, (\mathbf{y})^5, \mathbf{y}, Q_1, \mathbf{z}, (\mathbf{z})^5, Q_3, (\mathbf{e})^5, \mathbf{e} \rangle$.

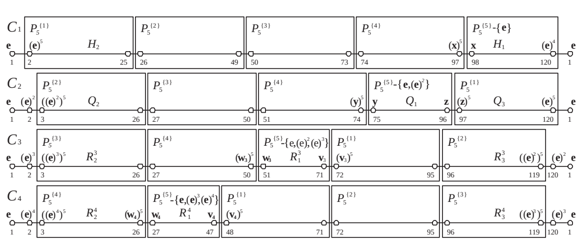

Figure 3: Illustration for Theorem 3 on P_5.

(**3**) $3 \leq k \leq 4$. By Lemma 6, there is a hamiltonian path R_1^k of $P_5^{\{b_{k,6-k}\}} - \{\mathbf{e}, (\mathbf{e})^{k-1}, (\mathbf{e})^k\}$ joining a node $\mathbf{w_k}$ with $(\mathbf{w_k})_1 = 4$ to a node $\mathbf{v_k}$ with $(\mathbf{v_k})_1 = 1$. By Theorem 2, there is a hamiltonian path R_2^k of $\cup_{t=1}^{5-k} P_5^{\{b_{k,t}\}}$ joining the node $((\mathbf{e})^k)^5$ to the node $(\mathbf{w_k})^5$ such that $R_2^k(i + 24(j-1)) \in P_5^{\{b_{k,j}\}}$ for every $i \in \langle 24 \rangle$ and for every $j \in \langle 5-k \rangle$. Again, there is a hamiltonian path R_3^k of $\cup_{t=7-k}^{5} P_5^{\{b_{k,t}\}}$ joining the node $(\mathbf{v_k})^5$ to the node $((\mathbf{e})^{k-1})^5$ such that $R_3^k(i + 24(j-1)) \in P_5^{\{b_{k,6-k+j}\}}$ for every $i \in \langle 24 \rangle$ and for every $j \in \langle k-1 \rangle$. We set $C_k = \langle \mathbf{e}, (\mathbf{e})^k, ((\mathbf{e})^k)^5, R_2^k, (\mathbf{w_k})^5, \mathbf{w_k}, R_1^k, \mathbf{v_k}, (\mathbf{v_k})^5, R_3^k, ((\mathbf{e})^{k-1})^5, (\mathbf{e})^{k-1}, \mathbf{e} \rangle$.

Then $\{C_1, C_2, C_3, C_4\}$ forms a set of 4-mutually independent hamiltonian cycles of P_5 starting from the node $\mathbf{e}$. See Figure 3 for illustration.

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